

DM-Inc ERP Hosting Recommendation

Recommended AWS production layout for the DM-Inc ERP, with cost minimisation as the primary objective. The design keeps the application stack close to the current codebase, keeps PostgreSQL lean, and uses a cost model in which the 5 TB document repository is held in Amazon S3 rather than inside a relational database.

EC2 + RDS PostgreSQL

5 TB document repository in S3

Single app node at launch

Graviton cost efficiency

Redis on-box initially

CURRENT LAUNCH MODEL

\$437.06

Estimated monthly total for the cost-optimised production launch configuration.

RESILIENCE UPGRADE

+\$197.47

Estimated monthly uplift to move the database tier from **Single-AZ** to **Multi-AZ**.

DOCUMENT STORAGE COST

\$117.76

Estimated monthly cost for **5,120 GB** of S3 Standard storage.

WHY S3 MATTERS

5x lower

5 TB in S3 Standard is far cheaper than carrying the same document repository in RDS gp3 storage.

Why this is the right commercial shape

- It keeps the stack close to the system as already built and documented.
- It avoids paying container-platform or Aurora complexity costs too early.
- It keeps the application tier inexpensive and easy to reason about operationally.
- It moves the 5 TB document estate into the cheapest appropriate storage tier rather than inflating the database bill.

Decision framing

The hosting recommendation is intentionally conservative at launch: keep the application tier simple, keep PostgreSQL focused on structured data, and move the binary document repository into S3.

Recommended architecture

LAYER	RECOMMENDED SERVICE	FIT TO CURRENT BUILD
Ingress / TLS	Application Load Balancer + ACM	The deployment docs already allow ALB in front of Nginx, including health checks and trusted proxy settings.
Application tier	1 x EC2 m7g.large in an Auto Scaling Group Desired 1, Min 1, Max 2	Matches the repo's existing Ubuntu deployment shape: Nginx + Gunicorn + worker + scheduler.
App host services	Nginx, Gunicorn, Redis, worker, scheduler on the app node	This is exactly how the current deployment assets are structured.
Relational database	Amazon RDS for PostgreSQL db.r7g.large	The codebase requires PostgreSQL in production and explicitly rejects SQLite for production use.
Document storage	Amazon S3 for PDFs, scans, uploaded files, generated exports	Important: S3-backed document storage should be implemented before go-live if the 5 TB document estimate is firm.
Operations	CloudWatch, SSM Session Manager, automated RDS backups, monthly snapshots	Fits the existing operational model and reduces administration overhead.

Codebase alignment with the current build

- **PostgreSQL is already the required production database.**
- **Multi-worker production already expects Redis.**
- **The deployment model already assumes Nginx + Gunicorn + worker + scheduler on Ubuntu/EC2.**
- **ALB in front of Nginx is already accounted for in deployment settings.**
- **S3 for document binaries is not yet a native current-state storage backend.**

What we are deliberately not buying on day one

- No ECS/Fargate container platform.
- No Aurora premium over standard RDS PostgreSQL.
- No separate ElastiCache Redis tier initially.
- No oversized app fleet before actual CPU, memory, queue, or availability pressure justifies it.
- No 5 TB PostgreSQL storage profile for document binaries.

COST VIEW

Monthly line-item estimate (USD)

COMPONENT	ASSUMPTION	RATE BASIS	ESTIMATED MONTHLY COST	COMMENT
Application Load Balancer	1 ALB, 730 hours/month	\$0.0225/hour	\$16.42	Fixed ALB hourly charge.
ALB usage allowance	1 average LCU, 730 hours/month	\$0.008/LCU-hour	\$5.84	Budgetary placeholder for light to moderate application traffic.
EC2 application node	1 x m7g.large , 730 hours/month	\$0.0816/hour	\$59.57	2 vCPU / 8 GiB Graviton app node.
EBS root/app volume	100 GB gp3	\$0.08/GB-month	\$8.00	Boot volume and local runtime storage only. Not used for the 5 TB document estate.
RDS PostgreSQL compute	1 x db.r7g.large , Single-AZ	\$0.239/hour	\$174.47	2 vCPU / 16 GiB memory-optimised RDS instance.
RDS PostgreSQL storage	200 GB gp3	\$0.115/GB-month	\$23.00	Assumes PostgreSQL stores structured application data only, not the document estate.
Amazon S3 document storage	5,120 GB S3 Standard	\$0.023/GB-month	\$117.76	Main storage line item for legal documents, scans, uploads, and generated binaries.
S3 requests / lifecycle allowance	Light request volume and simple lifecycle rules	Budgetary allowance	\$5.00	Small contingency for PUT/GET/list operations and lifecycle transitions.

COMPONENT	ASSUMPTION	RATE BASIS	ESTIMATED MONTHLY COST	COMMENT
CloudWatch + SSM	Basic logs, alarms, and Session Manager	Budgetary allowance	\$12.00	Operational visibility and managed host access.
Backup / snapshot allowance	Monthly manual snapshots and modest retained overhead	Budgetary allowance	\$15.00	Allows for retained backup overhead beyond the free automated backup baseline.
Estimated monthly total: cost-optimised launch configuration			\$437.06	Single app node + Single-AZ RDS + 5 TB document estate in S3.

Assumption set: US East (N. Virginia), on-demand pricing, 730-hour month, one application node, one database node, 200 GB PostgreSQL gp3 storage, and 5,120 GB of S3 Standard document storage.

DECISION VIEW

Optional resilience upgrade

CHANGE	SINGLE-AZ	MULTI-AZ	MONTHLY UPLIFT
RDS compute	\$174.47	\$348.94	+\$174.47
RDS storage (200 GB gp3)	\$23.00	\$46.00	+\$23.00
Revised monthly total	\$437.06	\$634.53	+\$197.47

Commercial interpretation: if DM-Inc wants a stronger database availability posture from day one, the approximate monthly uplift is just under \$200.

Why S3 changes the economics

If 5 TB were carried in PostgreSQL gp3 instead of S3:

RDS Single-AZ storage alone would be about **\$588.80/month** at \$0.115/GB-month, before database compute is added.

The same 5 TB in S3 Standard is about **\$117.76/month** at \$0.023/GB-month.

That is why the commercial recommendation is to keep PostgreSQL for structured application data and move the binary document repository into object storage.

Decision recommendation

Recommended launch for cost control: single application node + Single-AZ RDS PostgreSQL + S3-backed document storage.

Recommended upgrade if availability requirements are stricter: keep the same app tier and switch the database tier to Multi-AZ, taking the monthly estimate from **\$437.06** to about **\$634.53**.

Budgetary estimate prepared March 16, 2026 using AWS us-east-1 public price files published March 12-13, 2026. Figures exclude material outbound internet transfer, heavy document retrieval traffic, and any future HA redesign of the application tier.